McGill University

Department of Mechanical Engineering

MECH 309
Numerical Methods in Mechanical Engineering

Data-Calibrated Local Weather Prediction

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1. Introduction

Weather forecasting plays a critical role in many engineering applications, such as energy management, transportation, infrastructure planning and environmental protection. Accurate short-term predictions of atmospheric conditions are particularly important in urban areas, like Montréal, where temperature variations and meteorological conditions have a massive impact on heating and cooling, the dispersion of pollutants, and overall air quality, all of which affect millions of individuals.

While large-scale numerical weather prediction models provide highly detailed forecasts, such models require large amounts of computational power and are not always suitable for fast, localized analysis. As such, simple, data-driven models that can capture patterns in atmospheric behavior are often valuable tools for many practical applications.

The objective of this project is to develop a data-driven forecasting model for near-surface air temperature in Montréal using historical hourly data. The model aims to predict future temperatures over a variety of forecast horizons while remaining simple and computationally efficient. To achieve this, the approach combines data acquisition, preprocessing and regression-based modeling with model parameters that are estimated from training data, and performance evaluated on a different set of validation data.

The effectiveness of the model is then assessed using standard error metrics, including the root mean square error (RMSE) and the mean absolute error (MAE), and compared against a baseline model.

2. Data Acquisition and Preprocessing:

Historical meteorological data for Montréal was obtained using the publicly available Open-Meteo archive API. Starter-code was used to obtain two distinct datasets to create the model. The code first retrieved raw hourly weather data over the specified time period, before subsequently processing it into a complete, uniformly sampled time series suitable for regression-based modeling.

The first dataset, used for model-training purposes, covered the period of January 1, 2018 to December 29th, 2024, and included data on near-surface air temperature, wind speed and direction, relative humidity, surface pressure, precipitation and cloud cover.

	T	W	RH	P	Prec	Cloud	sin_day	cos_day	sin_year	cos_year	T_lag1	T_lag3	T_lag12	T_plus1	T_plus3	T_plus6	T_plus12	T_plus24
2018-01-01 12:00	-23.3	18.2	68	1018.4	0	0	1.22E-16	-1	0.017202	0.999852	-24.2	-25.5	-23.1	-22.4	-20.9	-22	-25.4	-21.2
2018-01-01 13:00	-22.4	18.6	66	1017.6	0	0	-0.25882	-0.96593	0.017202	0.999852	-23.3	-25	-23.5	-21.5	-20.8	-22.5	-27.1	-19.8
2018-01-01 14:00	-21.5	18.3	65	1017	0	0	-0.5	-0.86603	0.017202	0.999852	-22.4	-24.2	-24.1	-20.9	-21.3	-22.5	-26.8	-18.6
2018-01-01 15:00	-20.9	16.9	64	1016.7	0	0	-0.70711	-0.70711	0.017202	0.999852	-21.5	-23.3	-24.4	-20.8	-22	-23.2	-28	-17.7
2018-01-01 16:00	-20.8	15.1	65	1016.6	0	0	-0.86603	-0.5	0.017202	0.999852	-20.9	-22.4	-24.6	-21.3	-22.5	-23.1	-27.1	-16.9
2018-01-01 17:00	-21.3	14.1	68	1016.5	0	0	-0.96593	-0.25882	0.017202	0.999852	-20.8	-21.5	-24.8	-22	-22.5	-24.1	-26	-16.2
2018-01-01 18:00	-22	11.3	70	1016.5	0	2	-1	-1.84E-16	0.017202	0.999852	-21.3	-20.9	-25	-22.5	-23.2	-25.4	-25	-15.8
2018-01-01 19:00	-22.5	9.8	71	1016.4	0	8	-0.96593	0.258819	0.017202	0.999852	-22	-20.8	-25.1	-22.5	-23.1	-27.1	-25.3	-14.9
2018-01-01 20:00	-22.5	7.6	71	1016.4	0	32	-0.86603	0.5	0.017202	0.999852	-22.5	-21.3	-25.1	-23.2	-24.1	-26.8	-25.5	-15.9
2018-01-01 21:00	-23.2	7.6	71	1016.2	0	81	-0.70711	0.707107	0.017202	0.999852	-22.5	-22	-25.5	-23.1	-25.4	-28	-25.2	-17
2018-01-01 22:00	-23.1	7.6	72	1016.3	0	79	-0.5	0.866025	0.017202	0.999852	-23.2	-22.5	-25	-24.1	-27.1	-27.1	-23.8	-18.6
2018-01-01 23:00	-24.1	7.6	73	1016.1	0	43	-0.25882	0.965926	0.017202	0.999852	-23.1	-22.5	-24.2	-25.4	-26.8	-26	-22.6	-17.9
2018-01-02 0:00	-25.4	7.2	74	1016.1	0	37	0	1	0.034398	0.999408	-24.1	-23.2	-23.3	-27.1	-28	-25	-21.2	-16.5
2018-01-02 1:00	-27.1	10.9	75	1016	0	58	0.258819	0.965926	0.034398	0.999408	-25.4	-23.1	-22.4	-26.8	-27.1	-25.3	-19.8	-17.5
2018-01-02 2:00	-26.8	9.4	75	1015.8	0	71	0.5	0.866025	0.034398	0.999408	-27.1	-24.1	-21.5	-28	-26	-25.5	-18.6	-16.1
2018-01-02 3:00	-28	10.6	75	1016.1	0	92	0.707107	0.707107	0.034398	0.999408	-26.8	-25.4	-20.9	-27.1	-25	-25.2	-17.7	-15.2
2018-01-02 4:00	-27.1	10.4	73	1016.2	0	90	0.866025	0.5	0.034398	0.999408	-28	-27.1	-20.8	-26	-25.3	-23.8	-16.9	-17.1
2018-01-02 5:00	-26	8.6	74	1016.2	0	88	0.965926	0.258819	0.034398	0.999408	-27.1	-26.8	-21.3	-25	-25.5	-22.6	-16.2	-14.8
2018-01-02 6:00	-25	9	73	1016.1	0	57	1	6.12E-17	0.034398	0.999408	-26	-28	-22	-25.3	-25.2	-21.2	-15.8	-14.9
2018-01-02 7:00	-25.3	10.1	74	1016	0	16	0.965926	-0.25882	0.034398	0.999408	-25	-27.1	-22.5	-25.5	-23.8	-19.8	-14.9	-14.6
2018-01-02 8:00	-25.5	11	76	1016.6	0	97	0.866025	-0.5	0.034398	0.999408	-25.3	-26	-22.5	-25.2	-22.6	-18.6	-15.9	-15.1
2018-01-02 9:00	-25.2	11.8	78	1016.4	0	81	0.707107	-0.70711	0.034398	0.999408	-25.5	-25	-23.2	-23.8	-21.2	-17.7	-17	-15.8
2018-01-02 10:00	-23.8	12	78	1016.1	0	97	0.5	-0.86603	0.034398	0.999408	-25.2	-25.3	-23.1	-22.6	-19.8	-16.9	-18.6	-15.4
2018-01-02 11:00	-22.6	10.7	76	1015	0	100	0.258819	-0.96593	0.034398	0.999408	-23.8	-25.5	-24.1	-21.2	-18.6	-16.2	-17.9	-13.5
2018-01-02 12:00	-21.2	9.7	74	1013.9	0	100	1.22E-16	-1	0.034398	0.999408	-22.6	-25.2	-25.4	-19.8	-17.7	-15.8	-16.5	-10.3
2018-01-02 13:00	-19.8	7.3	70	1012.2	0.1	100	-0.25882	-0.96593	0.034398	0.999408	-21.2	-23.8	-27.1	-18.6	-16.9	-14.9	-17.5	-7.7
2018-01-02 14:00	-18.6	8.2	70	1010.3	0.2	100	-0.5	-0.86603	0.034398	0.999408	-19.8	-22.6	-26.8	-17.7	-16.2	-15.9	-16.1	-6.7
2018-01-02 15:00	-17.7	6.7	71	1009.3	0.3	100	-0.70711	-0.70711	0.034398	0.999408	-18.6	-21.2	-28	-16.9	-15.8	-17	-15.2	-7.1
2018-01-02 16:00	-16.9	3.4	73	1008.4	0.3	100	-0.86603	-0.5	0.034398	0.999408	-17.7	-19.8	-27.1	-16.2	-14.9	-18.6	-17.1	-7.8
2018-01-02 17:00	-16.2	3.9	77	1007.7	0.3	99	-0.96593	-0.25882	0.034398	0.999408	-16.9	-18.6	-26	-15.8	-15.9	-17.9	-14.8	-8.1
2018-01-02 18:00	-15.8	4.6	83	1007.4	0.2	99	-1	-1.84E-16	0.034398	0.999408	-16.2	-17.7	-25	-14.9	-17	-16.5	-14.9	-8.1
2018-01-02 19:00	-14.9	5.4	83	1007.2	0.2	100	-0.96593	0.258819	0.034398	0.999408	-15.8	-16.9	-25.3	-15.9	-18.6	-17.5	-14.6	-8.4
2018-01-02 20:00	-15.9	2.3	82	1007.8	0.1	99	-0.86603	0.5	0.034398	0.999408	-14.9	-16.2	-25.5	-17	-17.9	-16.1	-15.1	-8.9
2018-01-02 21:00	-17	5	81	1007.8	0	97	-0.70711	0.707107	0.034398	0.999408	-15.9	-15.8	-25.2	-18.6	-16.5	-15.2	-15.8	-9.1

Figure 1: Sample of Processed Training Data Set

The dataset also included additional parameters to better model temporal patterns. First, daily variations were represented using sin_day and cos_day, which account for daily cycles associated with day-night changes. Yearly variations were represented using sin_year and cos_year, which account for cyclical variations throughout the year, such as seasons. These parameters allowed the model to account for cyclic behaviour through time without introducing discontinuities in time representation, and come from:

$$\sin\left(\frac{2\pi t}{24}\right); \cos\left(\frac{2\pi t}{24}\right)$$

where t is the hour of the day, and:

$$\sin\left(\frac{2\pi d}{365.25}\right); \cos\left(\frac{2\pi d}{365.25}\right)$$

where d is the day of the year.

Lagged temperature terms T_lag1, T_lag3 and T_lag12 correspond to the temperature measured 1, 3 and 12 hours. Target values T_plus1, T_plus3, Tplus6, T_plus12 and T_plus24 represent the temperature 1, 3, 6 , 12 and 24 hours into the future, and were used as response variables the model was trained to estimate for each forecast horizon.

A second dataset covering the period of December 30th 2024 to December 29th 2025 was created using the same procedure. This dataset was reserved for validating the performance of the model.

	T	W	RH	P	Prec	Cloud	sin_day	cos_day	sin_year	cos_year	T_lag1	T_lag3	T_lag12	T_plus1	T_plus3	T_plus6	T_plus12	T_plus24
2024-12-30 0:00	6.9	19.2	95	995	0	100	0	1	-0.0043	0.999991	6.8	4.7	1.2	7.2	8.2	8.4	10.1	3.2
2024-12-30 1:00	7.2	20.6	95	993.9	0	100	0.258819	0.965926	-0.0043	0.999991	6.9	6.2	2	7.5	8.4	8.6	9.9	2.8
2024-12-30 2:00	7.5	22.4	96	992.6	0	100	0.5	0.866025	-0.0043	0.999991	7.2	6.8	2.2	8.2	8	8.6	8.1	2.6
2024-12-30 3:00	8.2	21.2	94	991.6	0.4	100	0.707107	0.707107	-0.0043	0.999991	7.5	6.9	2.2	8.4	8.4	10.1	8.2	2.7
2024-12-30 4:00	8.4	25.5	93	991.2	0.4	100	0.866025	0.5	-0.0043	0.999991	8.2	7.2	2.9	8	8.6	10.1	7.9	2.5
2024-12-30 5:00	8	23	95	990.6	0.1	100	0.965926	0.258819	-0.0043	0.999991	8.4	7.5	2.5	8.4	8.6	9.2	6.1	2.5
2024-12-30 6:00	8.4	23.9	94	989.8	0	100	1	6.12E-17	-0.0043	0.999991	8	8.2	1.9	8.6	10.1	10.1	5.3	2.5
2024-12-30 7:00	8.6	25.4	93	988.8	0	100	0.965926	-0.25882	-0.0043	0.999991	8.4	8.4	1.9	8.6	10.1	9.9	4.7	2.5
2024-12-30 8:00	8.6	24.4	93	988.4	0	100	0.866025	-0.5	-0.0043	0.999991	8.6	8	2.2	10.1	9.2	8.1	4.3	2.5
2024-12-30 9:00	10.1	23.7	90	986.7	0	100	0.707107	-0.70711	-0.0043	0.999991	8.6	8.4	4.7	10.1	10.1	8.2	4.4	3.2
2024-12-30 10:00	10.1	20.5	90	987	0.5	100	0.5	-0.86603	-0.0043	0.999991	10.1	8.6	6.2	9.2	9.9	7.9	4.1	3.2
2024-12-30 11:00	9.2	16.7	91	987	0.1	100	0.258819	-0.96593	-0.0043	0.999991	10.1	8.6	6.8	10.1	8.1	6.1	3.7	3.6
2024-12-30 12:00	10.1	10.8	88	986.6	0.3	99	1.22E-16	-1	-0.0043	0.999991	9.2	10.1	6.9	9.9	8.2	5.3	3.2	4.2
2024-12-30 13:00	9.9	13.3	89	986.7	0.4	100	-0.25882	-0.96593	-0.0043	0.999991	10.1	10.1	7.2	8.1	7.9	4.7	2.8	4.2
2024-12-30 14:00	8.1	14.6	83	986.5	0	100	-0.5	-0.86603	-0.0043	0.999991	9.9	9.2	7.5	8.2	6.1	4.3	2.6	4.3
2024-12-30 15:00	8.2	16.6	76	986.2	0	100	-0.70711	-0.70711	-0.0043	0.999991	8.1	10.1	8.2	7.9	5.3	4.4	2.7	4.9
2024-12-30 16:00	7.9	14.9	76	986.8	0	100	-0.86603	-0.5	-0.0043	0.999991	8.2	9.9	8.4	6.1	4.7	4.1	2.5	4.7
2024-12-30 17:00	6.1	16.4	76	987.6	0	100	-0.96593	-0.25882	-0.0043	0.999991	7.9	8.1	8	5.3	4.3	3.7	2.5	2.8
2024-12-30 18:00	5.3	16.7	73	988.5	0	95	-1	-1.84E-16	-0.0043	0.999991	6.1	8.2	8.4	4.7	4.4	3.2	2.5	1.1
2024-12-30 19:00	4.7	18.7	77	989	0.1	100	-0.96593	0.258819	-0.0043	0.999991	5.3	7.9	8.6	4.3	4.1	2.8	2.5	2.2
2024-12-30 20:00	4.3	18.9	82	989.7	0.2	100	-0.86603	0.5	-0.0043	0.999991	4.7	6.1	8.6	4.4	3.7	2.6	2.5	2.4
2024-12-30 21:00	4.4	19.8	85	989.8	0	100	-0.70711	0.707107	-0.0043	0.999991	4.3	5.3	10.1	4.1	3.2	2.7	3.2	1.5
2024-12-30 22:00	4.1	21.6	86	990	0	100	-0.5	0.866025	-0.0043	0.999991	4.4	4.7	10.1	3.7	2.8	2.5	3.2	1.5
2024-12-30 23:00	3.7	21.4	87	990.7	0.1	100	-0.25882	0.965926	-0.0043	0.999991	4.1	4.3	9.2	3.2	2.6	2.5	3.6	1.4
2024-12-31 0:00	3.2	21	89	991.3	0.1	100	0	1	0.012901	0.999917	3.7	4.4	10.1	2.8	2.7	2.5	4.2	1.2
2024-12-31 1:00	2.8	20.9	89	991.9	0.1	100	0.258819	0.965926	0.012901	0.999917	3.2	4.1	9.9	2.6	2.5	2.5	4.2	0.9
2024-12-31 2:00	2.6	21.6	90	992.6	0	100	0.5	0.866025	0.012901	0.999917	2.8	3.7	8.1	2.7	2.5	2.5	4.3	0.3
2024-12-31 3:00	2.7	20.8	89	993.6	0	99	0.707107	0.707107	0.012901	0.999917	2.6	3.2	8.2	2.5	2.5	3.2	4.9	0.4
2024-12-31 4:00	2.5	19.9	89	994.6	0	100	0.866025	0.5	0.012901	0.999917	2.7	2.8	7.9	2.5	2.5	3.2	4.7	0
2024-12-31 5:00	2.5	19.2	89	995.3	0	100	0.965926	0.258819	0.012901	0.999917	2.5	2.6	6.1	2.5	2.5	3.6	2.8	0
2024-12-31 6:00	2.5	18.6	89	996.3	0	100	1	6.12E-17	0.012901	0.999917	2.5	2.7	5.3	2.5	3.2	4.2	1.1	0.2
2024-12-31 7:00	2.5	18.2	88	997.3	0	100	0.965926	-0.25882	0.012901	0.999917	2.5	2.5	4.7	2.5	3.2	4.2	2.2	0.4
2024-12-31 8:00	2.5	16.6	88	998.2	0	100	0.866025	-0.5	0.012901	0.999917	2.5	2.5	4.3	3.2	3.6	4.3	2.4	0.7
2024-12-31 9:00	3.2	14.4	86	999	0	100	0.707107	-0.70711	0.012901	0.999917	2.5	2.5	4.4	3.2	4.2	4.9	1.5	0.6

Figure 2: Sample of Processed Validation Dataset

During the data-acquisition process, it was essential to keep the training and validation datasets strictly separate to ensure a reliable evaluation of the model's performance. The model parameters were estimated using only the training data, allowing the model to 'learn' parameters without being exposed to the validation data. If this were not the case, the model could become biased and represent predictions much more accurately than it otherwise would, which would not match its true predictive capability.

3. Model Formulation:

3.1: Predictor Variables and Design Matrix:

The final model was chosen to be a linear model of the form:

$$y = Bx,$$

where y is the forecasted temperatures in the form:

$$y = \begin{bmatrix} T_{+1} \\ T_{+3} \\ T_{+6} \\ T_{+12} \\ T_{+24} \end{bmatrix},$$

B is the matrix of coefficients which weight the values of the predictor variables and has the form:

$$B = \begin{bmatrix} \beta_{1,1} & \beta_{1,2} & \beta_{1,3} & \beta_{1,4} & \beta_{1,5} & \beta_{1,6} & \beta_{1,7} & \beta_{1,8} & \beta_{1,9} & \beta_{1,10} & \beta_{1,11} & \beta_{1,12} & \beta_{1,13} & \beta_{1,14} \\ \beta_{2,1} & \beta_{2,2} & \beta_{2,3} & \beta_{2,4} & \beta_{2,5} & \beta_{2,6} & \beta_{2,7} & \beta_{2,8} & \beta_{2,9} & \beta_{2,10} & \beta_{2,11} & \beta_{2,12} & \beta_{2,13} & \beta_{2,14} \\ \beta_{3,1} & \beta_{3,2} & \beta_{3,3} & \beta_{3,4} & \beta_{3,5} & \beta_{3,6} & \beta_{3,7} & \beta_{3,8} & \beta_{3,9} & \beta_{3,10} & \beta_{3,11} & \beta_{3,12} & \beta_{3,13} & \beta_{3,14} \\ \beta_{4,1} & \beta_{4,2} & \beta_{4,3} & \beta_{4,4} & \beta_{4,5} & \beta_{4,6} & \beta_{4,7} & \beta_{4,8} & \beta_{4,9} & \beta_{4,10} & \beta_{4,11} & \beta_{4,12} & \beta_{4,13} & \beta_{4,14} \\ \beta_{5,1} & \beta_{5,2} & \beta_{5,3} & \beta_{5,4} & \beta_{5,5} & \beta_{5,6} & \beta_{5,7} & \beta_{5,8} & \beta_{5,9} & \beta_{5,10} & \beta_{5,11} & \beta_{5,12} & \beta_{5,13} & \beta_{5,14} \end{bmatrix}$$

and x is the vector of input predictor variables used with the matrix of coefficients and is of the form:

$$x = \begin{bmatrix} 1 \\ T_{curr} \\ W \\ \sin\left(\frac{2\pi t}{24}\right) \\ \cos\left(\frac{2\pi t}{24}\right) \\ \sin\left(\frac{2\pi d}{365}\right) \\ \cos\left(\frac{2\pi d}{365}\right) \\ R_H \\ P_{pressure} \\ P_{precipitation} \\ C_{Cloud\ Coverage} \\ T_{Lag\ 1h} \\ T_{Lag\ 3h} \\ T_{Lag\ 12h} \end{bmatrix}$$

A linear model was chosen to predict the various temperature forecasts as the model sufficiently captures trends in the training data which it can use to predict future temperatures within a reasonable margin. The linear model is also much simpler in solving and interpreting compared to a more-complex non-linear model. This configuration reduces computing power and computing time significantly which allowed us to more efficiently assess the importance of different predictor variables when designing the model and assessing the effects of changing the data sampling.

A linear model also improved the feasibility of training the model under different conditions, to train the model a linear least squares approach was used of the form:

$$y_k = X\beta_k,$$

where y_k is the set of target temperatures, X is the matrix of predictor variables, and β_k is the vector of unknown weight coefficients of the final linear model B, such that the rows of B are comprised of the various β_k vectors trained for each forecast horizon.

$$B = \begin{bmatrix} \leftarrow & \beta_1 & \rightarrow \\ \leftarrow & \beta_2 & \rightarrow \\ \leftarrow & \beta_3 & \rightarrow \\ \leftarrow & \beta_4 & \rightarrow \\ \leftarrow & \beta_5 & \rightarrow \end{bmatrix}$$

The idea behind this form of training is that each forecast horizon will be trained separately to better estimate that forecasts trends, and the resulting combination of individual forecast horizons creates one larger model which predicts various forecast horizons given one set of predictor variables.

In training the model, it is apparent that the X matrix is tall, meaning that for the matrix of size $m \times n$, $m \gg n$. For a tall matrix of linear structure, the least squares method of solving for β_k is preferred. Other linear solving methods such as Gradient Descent and Newtons Method were considered, however for the large size of the training data being multiple years, direct least squares method was preferred given its computational simplicity and quickness in its ability to produce accurate results [1]. Overall, to produce the final prediction model, linear least squares solving is applied 5 times for the 5 different forecast horizons, and the results of each are combined into a single matrix which predicts 5 different forecast horizons given a single set of predictor variables.

Initially, a linear model with predictor variables: Current Temperature; Wind Speed; Time of Day (sin and cos); Intercept, and a 1 Hour Lag, was considered. This oversimplified model was used as a basis to set up the model in GitHub and Python and to initially attempt to observe how the model reacts to changing various parameters such as predictor variables and chosen training data across different forecast horizons. New predictor variables were chosen based on recommendations from the professor and [2] which explores the effects of various weather phenomena on temperature modelling. In adding new predictor variables to trial, two factors were considered: the error compared to the model's predecessor; and the weight of the predictor variables respective weight coefficient. If the predictor variable's effect on the error reduction was marginal (i.e. less than $0.001°C$ RMSE or MAE) or the weight of the predictor variable was near 0 (i.e. less than 1×10^{-5}), the predictor variable was deemed ineffective and not chosen to remain in the final model. This process was repeated iteratively until a result was achieved that was deemed sufficiently accurate compared to the baseline model and compared to previous iterations of itself, where the final set of predictor variables consist of the following:

Table 1: Predictor Variables

Symbol:	**Variable Name**	**Description:**
T	Temperature	Near-Surface air Temperature
W	Wind Speed	Near-Surface Wind Speed
RH	Relative Humidity	Near-Surface Relative Humidity
P	Pressure	Surface Atmospheric Pressure
Prec	Precipitation	Hourly Precipitation amount
Cloud	Cloud Cover	Cloud Cover
1	Intercept	Constant offset
sin_day	Daily Sine Term	Daily Cycle Sine Term
cos_day	Daily Cosine Term	Daily Cycle Cosine Term
sin_year	Yearly Sine Term	Yearly Cycle Sine Term
cos_year	Yearly Cosine Term	Yearly Cycle Cosine Term
T_lag1	1-Hour Temperature Lag	Temperature 1 hour earlier

T_lag3	3-Hour Temperature Lag	Temperature 3 hours earlier
T_lag12	12-Hour Temperature Lag	Temperature 12 hours earlier

Where t is the time of day and d is the day of year.

3.2: Estimated Model Parameters:

The constructed design matrix of the form $y = Bx$, using the training dataset from January 1st, 2018, to December 30th, 2024, results in the following matrix

$$B =$$

-7.9000701	1.3467169	-0.0008994	0.2174699	-0.3634418	-0.066667	-0.3285331	0.0098426	0.0072928	-0.0037257	0.0001883	-0.3143214	-0.0576258	0.0060954
-27.786296	1.5992292	-0.0072977	1.1099273	-1.0975569	-0.2489722	-1.1961822	0.0339272	0.0258098	-0.0176374	0.0002064	-0.606907	-0.0800932	0.0169901
-52.436762	1.453895	-0.024604	2.8911247	-1.160212	-0.5314558	-2.3950564	0.0593339	0.0492899	-0.1049115	0.0002716	-0.6194554	-0.0172834	0.0377939
-95.247401	1.0566255	-0.0560793	3.0705618	2.5767011	-0.8780275	-3.6082025	0.05951	0.0927704	-0.1534673	0.001634	-0.3269997	-0.0891382	0.1344854
-164.77562	0.8738639	-0.0771482	-1.0066135	-0.2619406	-1.7408901	-5.6555825	0.0186183	0.1663658	0.0797124	-0.0008402	-0.0671018	-0.1140007	-0.0679721

4. Results and Model Validation:

4.1 : Results and Validation Plots:

The predictions obtained from the model were plotted over the entire validation period (year of 2025) for each of the forecast horizons of 1, 3, 6, 12 and 24 hours. This allowed to evaluate the model's performance for different forecast horizons over a large set of data which spanned an entire yearly cycle. These predictions were also compared with the actual recorded weather data over the same period.

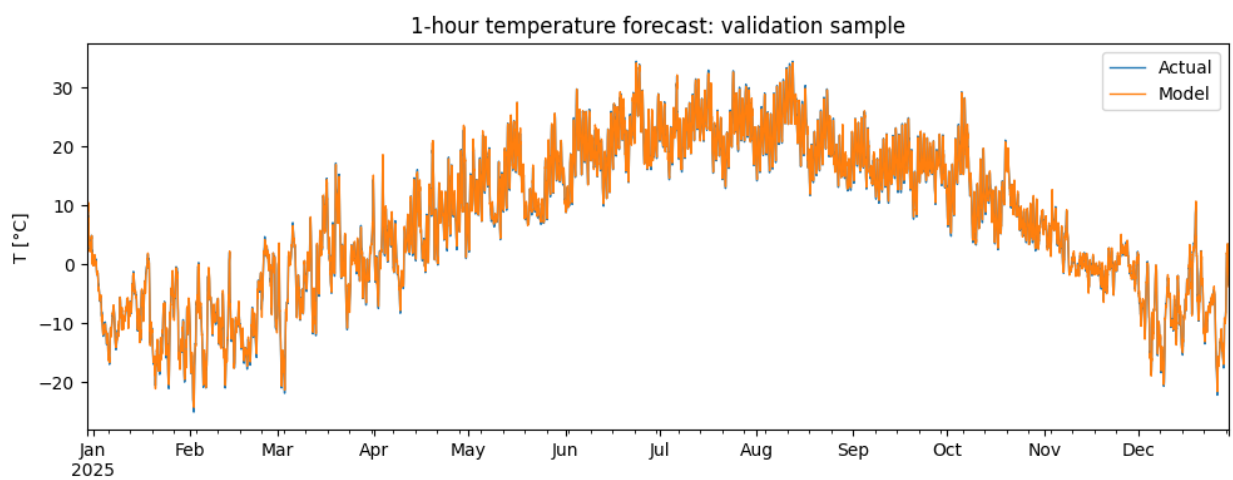

Figure 3: 1-Hour Temperature Forecast Validation Plot

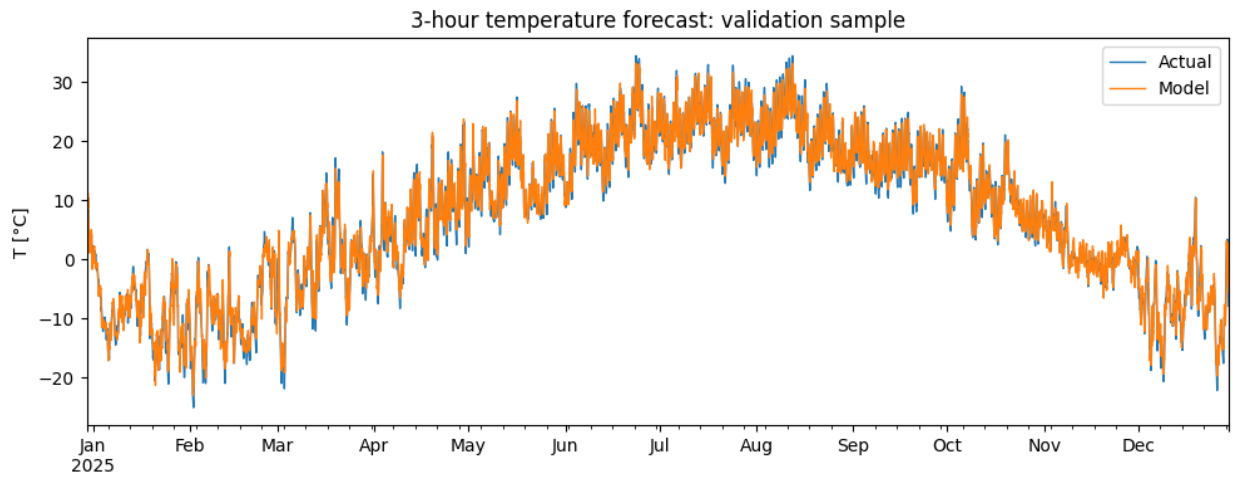

Figure 4: 3-Hour Temperature Forecast Validation Plot

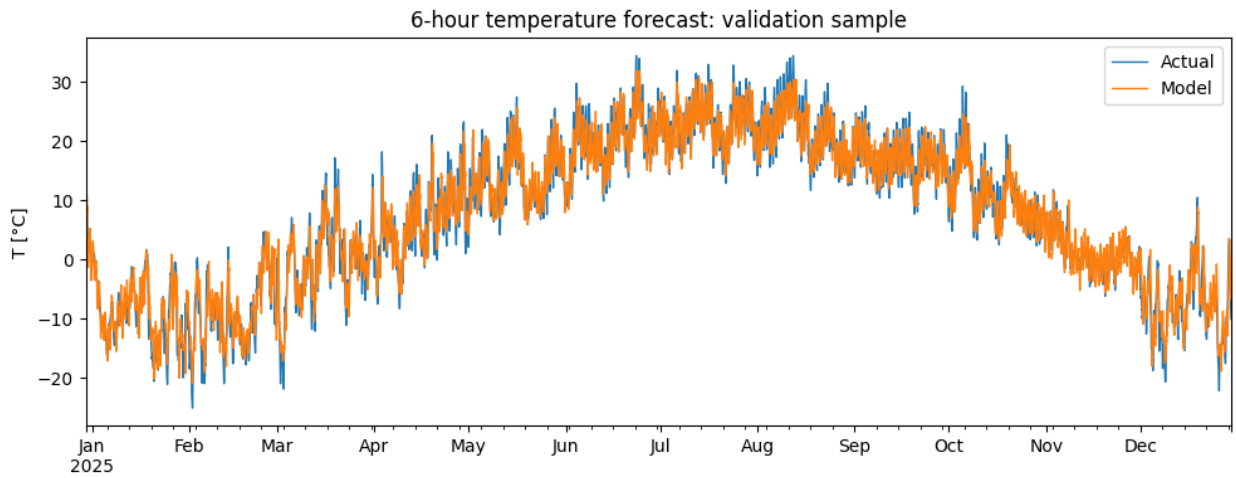

Figure 5: 6-Hour Temperature Forecast Validation Plot

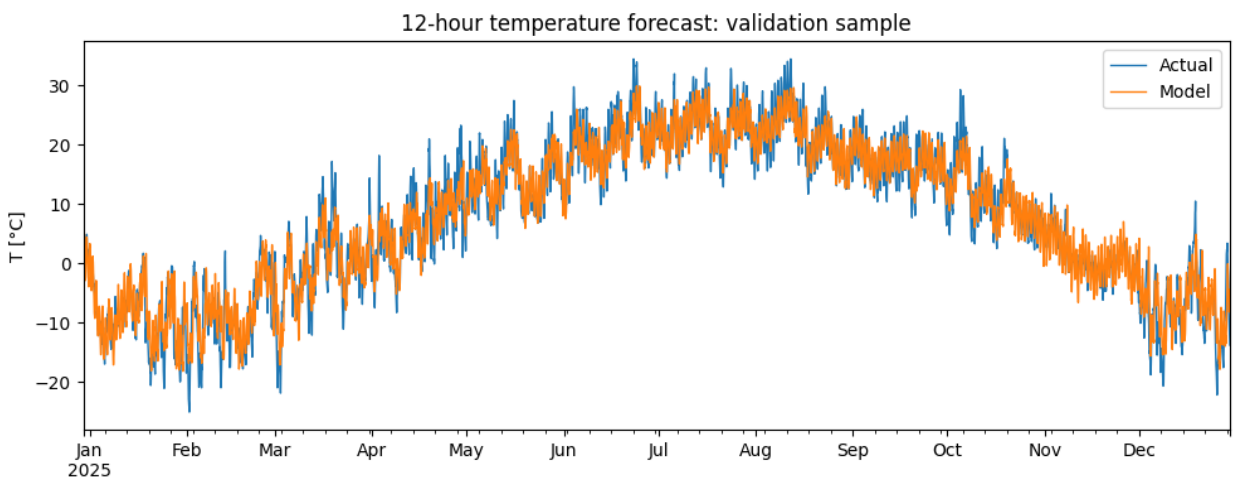

Figure 6: 12-Hour Temperature Forecast Validation Plot

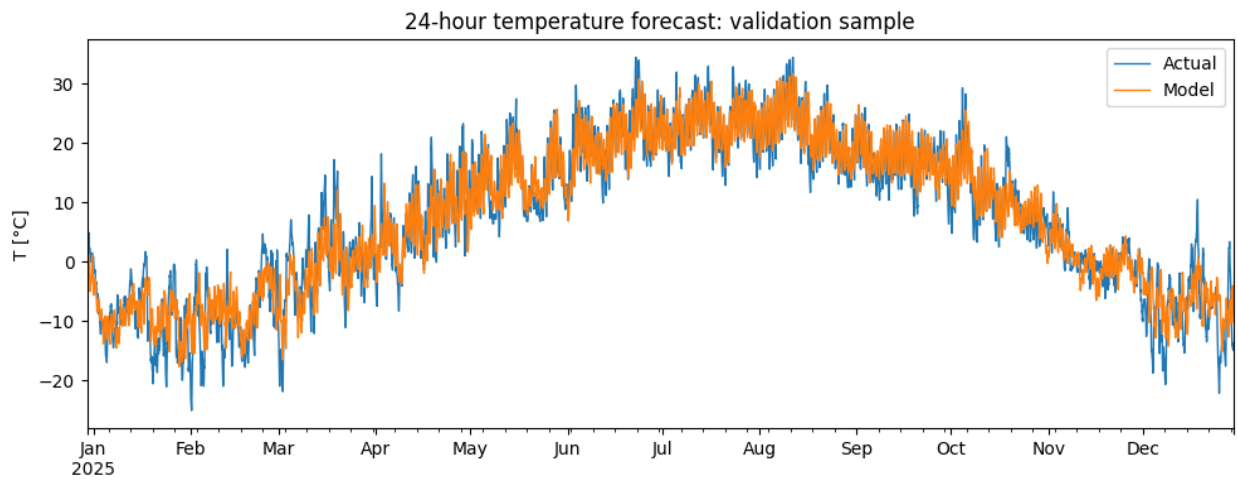

Figure 7: 24-Hour Temperature Forecast Validation Plot

Overall, the model was found to track the measured temperature trends well for all forecast horizons, with an expected increase in deviations for larger prediction horizons.

To evaluate if our model performed differently with seasons, further predictions were made specifically for winter and summer times. The results obtained show that in general, the model was more accurate in predicting the temperature forecasts in the summer months as opposed to the winter months. This is to be expected as predicting winter temperatures is often more challenging and can require more complex models that focus on different (more) predictor variables [3], [4]. In creating the model presented here, these complications were overlooked to create a more general application for relatively accurate use.

Table 2: RMSE in the Summer and Winter

RMSE	Summer	Winter
1h	0.623	0.608
3h	1.321	1.587
6h	1.964	2.824
12h	2.351	4.256
24h	2.634	4.915

Table 3: MAE in the Summer and Winter

MAE	Summer	Winter
1h	0.426	0.435
3h	1.006	1.214
6h	1.494	2.941
12h	1.797	3.690
24h	2.054	3.962

4.2: Error Metrics and Comparison with the Baseline Model:

The model was compared against a simple persistence model, which serves as a baseline reference. In this baseline approach, the temperature at the next time step is assumed to be the same as that of the current time step:

$$T_{k+h} \approx T_k$$

To quantitatively evaluate the model's performance, the root mean square error (RMSE) and the mean absolute error (MAE) were calculated to measure the difference between predicted and measured temperatures over the validation period. These values were also calculated for the baseline model, thus offering a benchmark to assess the performance of the regression model. It is important to note here that the error metrics presented in Table 4 apply to the entire validation period (2025 calendar year), as opposed to the results of Table 2, which apply only to the summer or winter months.

Table 4: Error Metrics for Regression and Baseline Models

Forecast Horizon	Model RMSE [°C]	Model MAE [°C]	Baseline RMSE [°C]	Baseline MAE [°C]
1h	0.623	0.439	0.896	0.676
3h	1.501	1.135	2.303	1.808
6h	2.460	1.870	3.968	3.172
12h	3.378	2.553	5.636	4.570
24h	3.800	2.876	4.838	3.556

Table 4 shows that the regression model outperformed the baseline model on all error metrics and for all time horizons. As expected, both the RMSE and the MAE increased for larger forecast horizons.

5. Discussion:

Analysis of the data-driven forecasting model showed consistent and accurate weather predictions over multiple time horizons, with an expected decrease in accuracy for larger time horizons. The model elaborated was simple, efficient and easy to interpret while still providing good predictive performance. The results also showed a clear edge in performance over a baseline simple persistence model, therefore demonstrating that the selected predictor variables provided meaningful information for short-term weather forecasting.

However, the model also had numerous limitations. First, it was based on a linear regression, which is not fully reflective of real-life atmospheric processes, which are often non-linear. As such, the model was fundamentally limited. Furthermore, our model relied only on local (Montreal) meteorological data, meaning that it did not account for larger scale atmospheric phenomena, such as pressure systems, weather fronts and more. Additionally, error metrics showed that our model was considerably less accurate at predicting weather during the winter months as opposed to the summer months, although this was likely because winter weather is inherently more unpredictable than summer weather, as well as to the limitations of a linear model. Finally, the model did not consider potential interactions of predictor variables with each other. This likely further contributed to a decrease in model accuracy, as more complex data

patterns were not accounted for. Finally, error metrics were shown to increase with forecast horizons which indicates a decrease in model performance as the forecast periods became longer, although this was expected and reflects the uncertainty of longer-range predictions.

Possible improvements to the model would include adding more predictor variables, such as wind direction or other lagged variables to further increase the accuracy of the model. Another improvement would be the implementation of non-linearity within the model, such as the use of a polynomial regression. Finally, the model could be improved by expanding the scope of the data used for training the model. This could be done, for example, by using data from neighbouring weather stations, which could therefore allow the model to consider larger-scale atmospheric phenomena.

However, the proposed improvements would likely make the model much more complex, which would result in an increase in necessary computational power. The currently developed model falls within the confines of the objective of this project, which was to create a simplified model for computationally light weather forecasting.

6. Conclusion:

In this project, a data-driven linear regression model was developed to predict near-surface air temperature in Montreal at forecast horizons of 1, 3, 6, 12 and 24 hours. Using historical hourly meteorological data, a set of measured and engineered predictor variables was constructed, which included, meteorological variables, cyclical parameters and lagged parameters. The model was then calibrated on a training dataset spanning January 1, 2018 to December 29th, 2024, before being evaluated on a separate dataset spanning December 30th 2024 to December 29th 2025. Root mean square error (RMSE) and mean absolute error (MAE) were computed over the summer months, winter months, and the entire validation time period, and the obtained values were compared with those from a baseline model spanning the same time periods.

The results showed that the regression model predicted the main temperature trends well and consistently outperformed the baseline for all forecast horizons, therefore demonstrating the effectiveness of our simple linear model. As expected, prediction errors increased with the forecast horizon, which can be attributed to greater uncertainty of longer-term temperature forecasts.

Overall, this project showed that relatively simple data-driven models are adequate short-term temperature predictors, while also being advantageous due to their simplicity, ease of interpretation, and low computational cost.

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